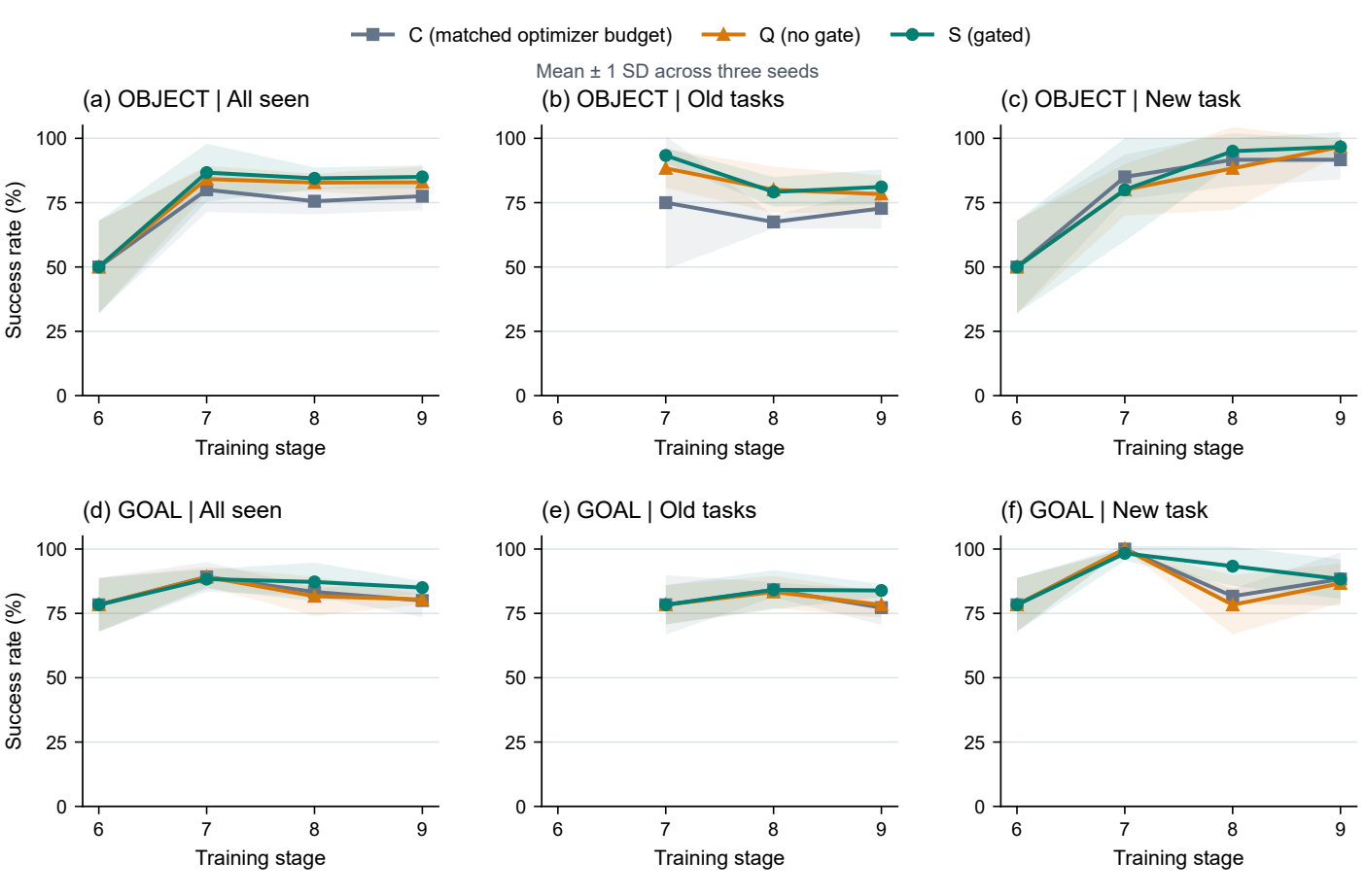
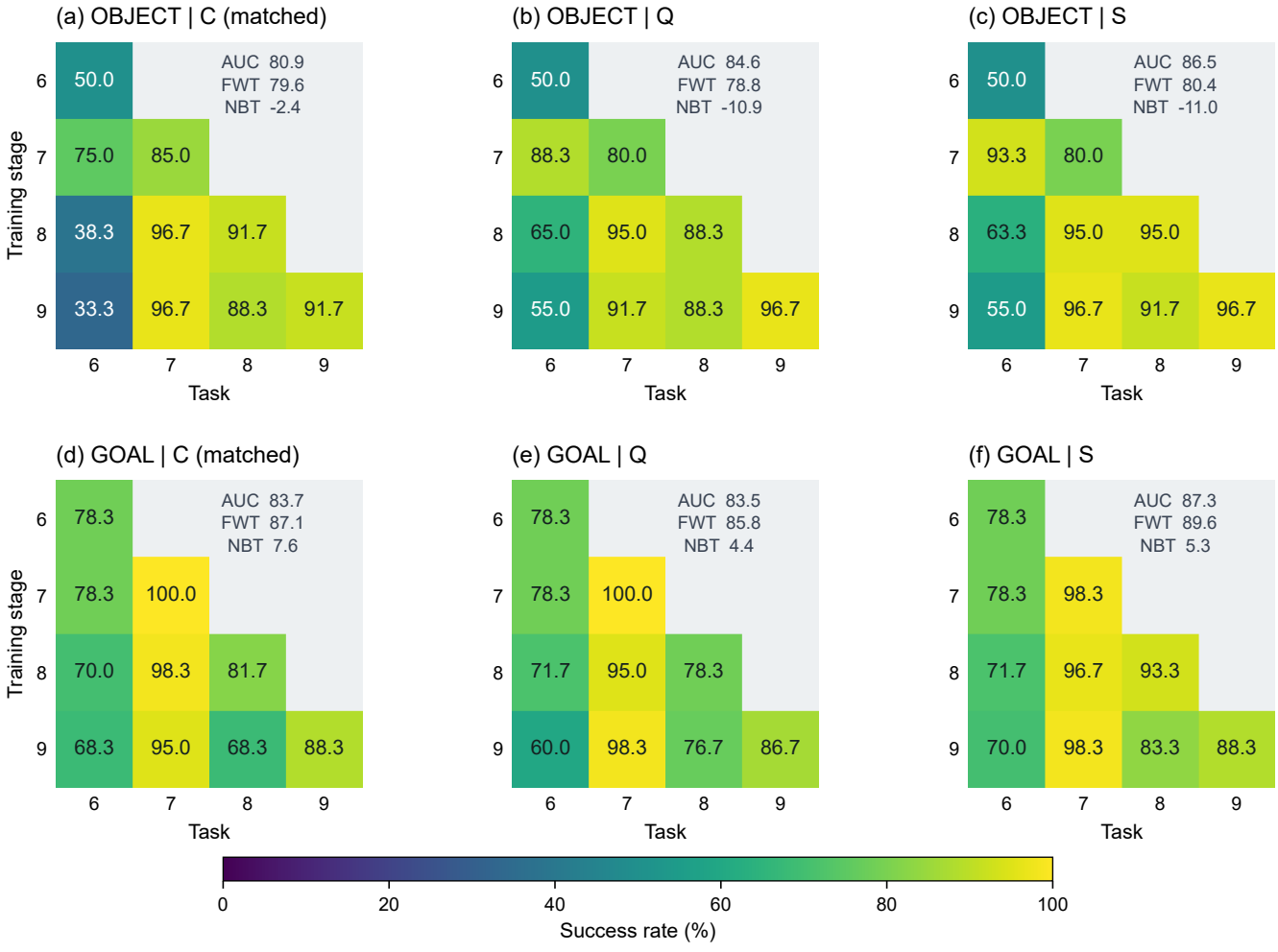
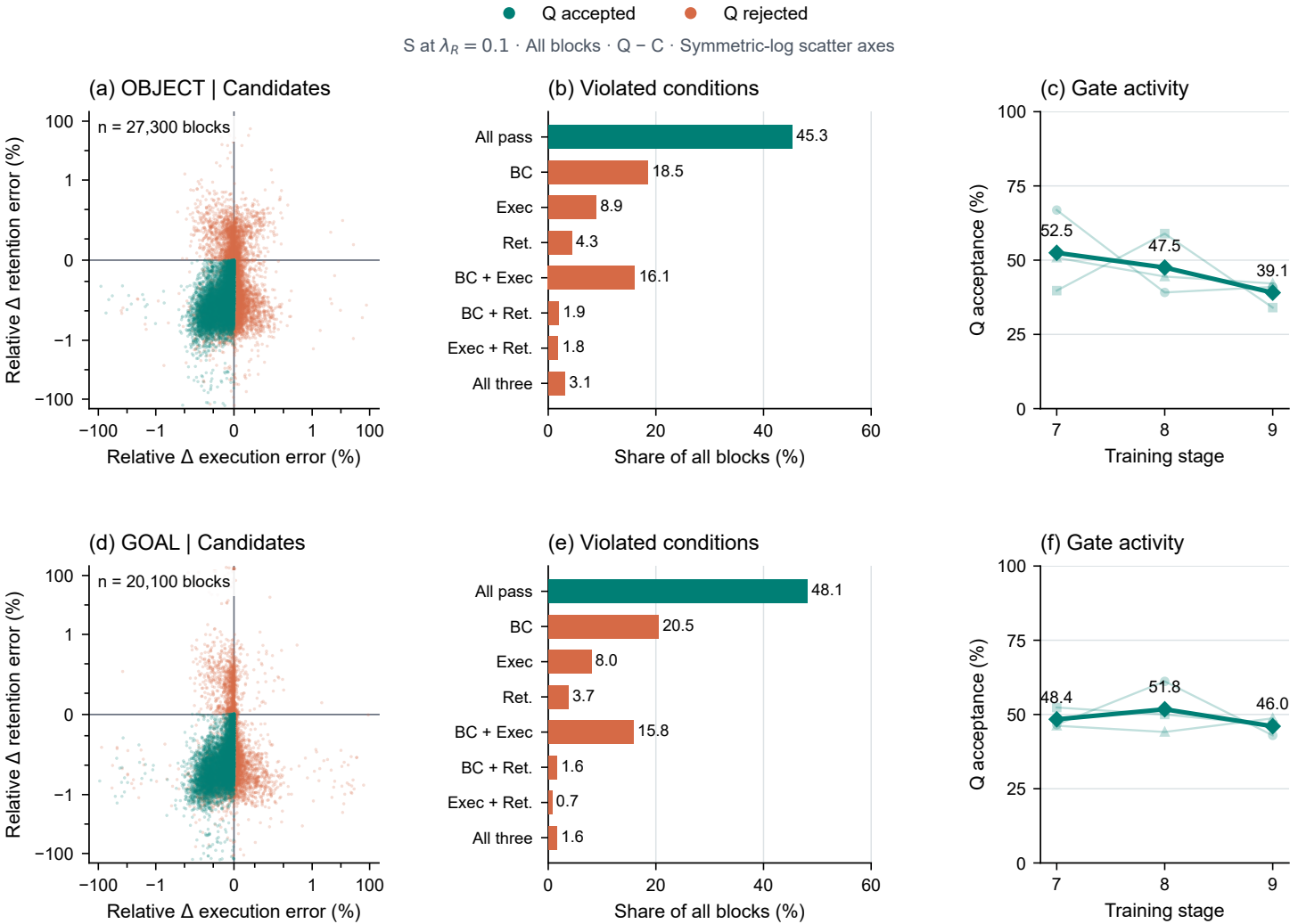
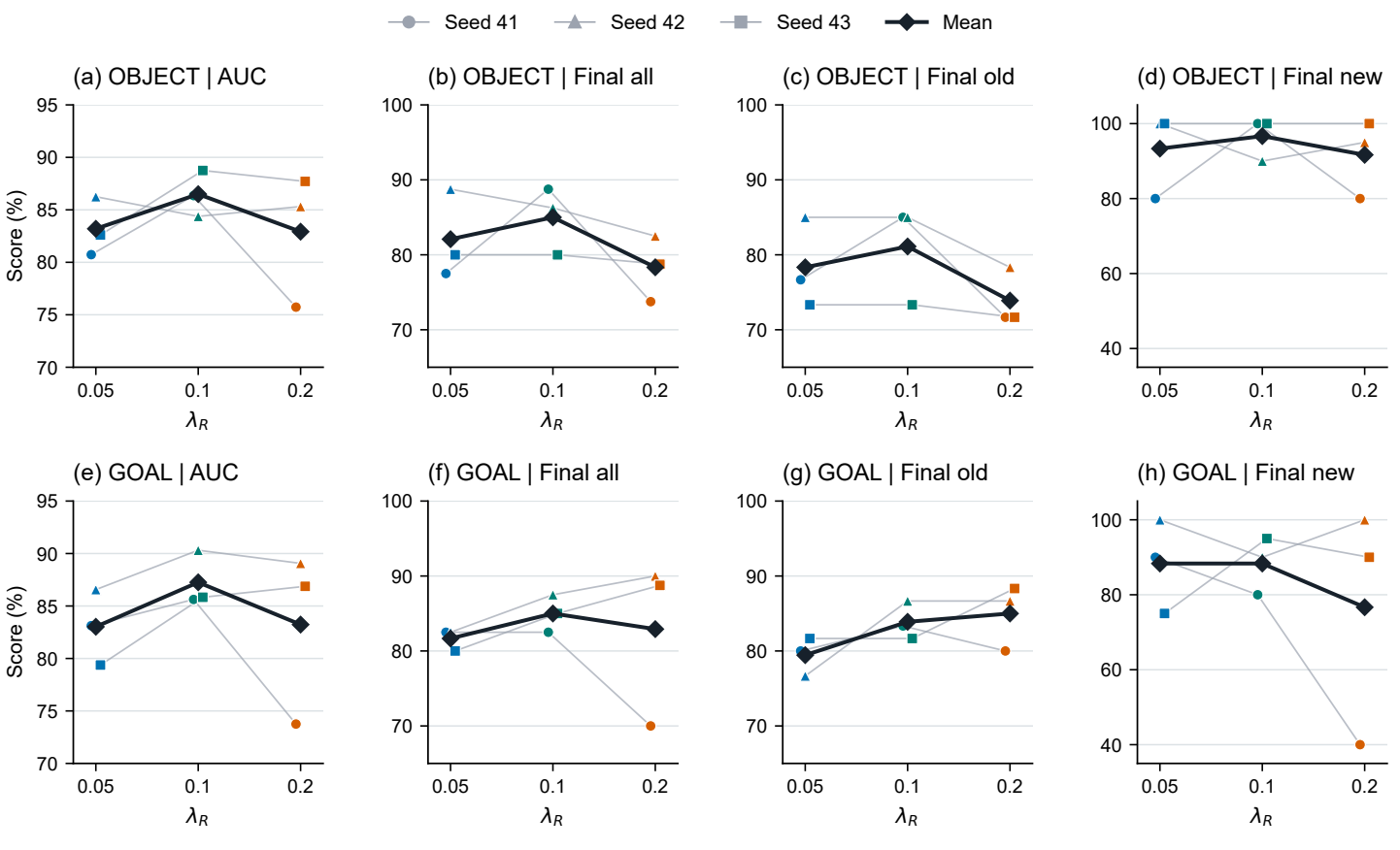
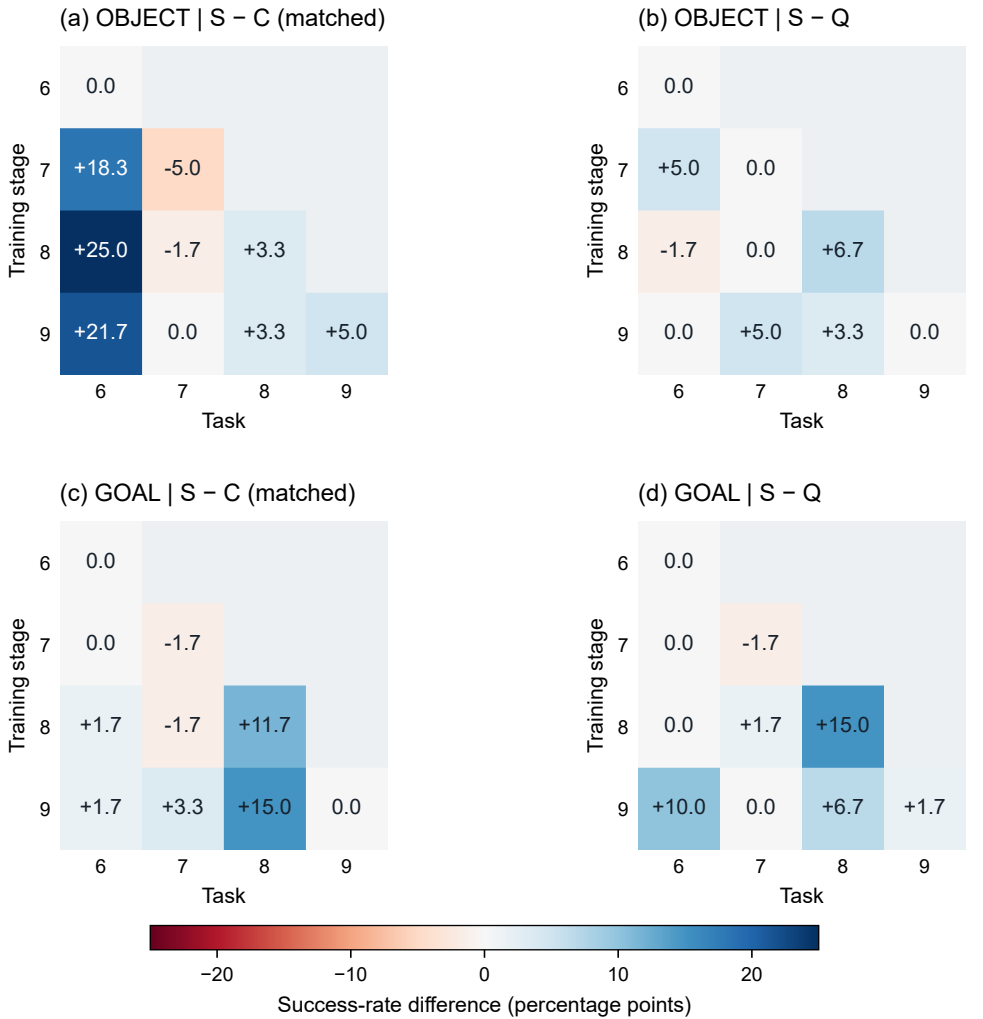
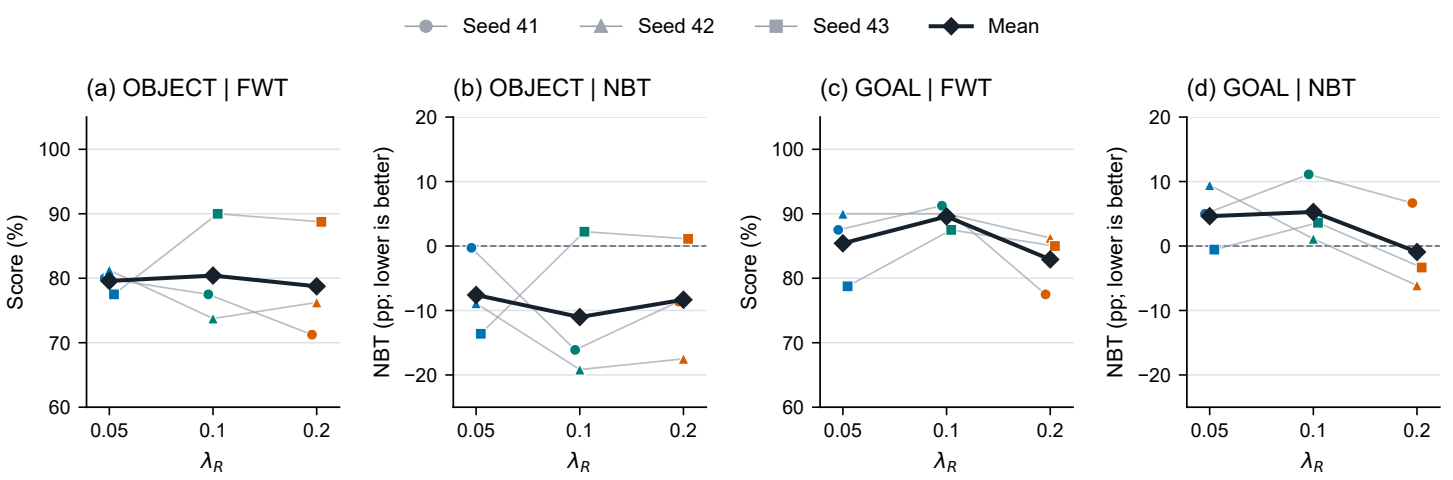

C (matched optimizer budget)
Q (no gate)
S (gated)
Mean ± 1 SD across three seeds
(a) OBJECT | All seen
(b) OBJECT | Old tasks
(c) OBJECT | New task
(d) GOAL | All seen
(e) GOAL | Old tasks
(f) GOAL | New task
Success rate (%)
Training stage
0
25
50
75
100
6
7
8
9


(a) OBJECT | C (matched)
AUC 80.9
FWT 79.6
NBT -2.4
(b) OBJECT | Q
AUC 84.6
FWT 78.8
NBT -10.9
(c) OBJECT | S
AUC 86.5
FWT 80.4
NBT -11.0
(d) GOAL | C (matched)
AUC 83.7
FWT 87.1
NBT 7.6
(e) GOAL | Q
AUC 83.5
FWT 85.8
NBT 4.4
(f) GOAL | S
AUC 87.3
FWT 89.6
NBT 5.3
Training stage
Task
Success rate (%)


Q accepted
Q rejected
S at $\lambda_R = 0.1$ · All blocks · Q − C · Symmetric-log scatter axes
(a) OBJECT | Candidates
n = 27,300 blocks
Relative Δ retention error (%)
Relative Δ execution error (%)
(b) Violated conditions
All pass
BC
Exec
Ret.
BC + Exec
BC + Ret.
Exec + Ret.
All three
45.3
18.5
8.9
4.3
16.1
1.9
1.8
3.1
Share of all blocks (%)
(c) Gate activity
Q acceptance (%)
52.5
47.5
39.1
Training stage
(d) GOAL | Candidates
n = 20,100 blocks
Relative Δ retention error (%)
Relative Δ execution error (%)
(e) Violated conditions
All pass
BC
Exec
Ret.
BC + Exec
BC + Ret.
Exec + Ret.
All three
48.1
20.5
8.0
3.7
15.8
1.6
0.7
1.6
Share of all blocks (%)
(f) Gate activity
Q acceptance (%)
48.4
51.8
46.0
Training stage


Seed 41
Seed 42
Seed 43
Mean
(a) OBJECT | AUC
(b) OBJECT | Final all
(c) OBJECT | Final old
(d) OBJECT | Final new
(e) GOAL | AUC
(f) GOAL | Final all
(g) GOAL | Final old
(h) GOAL | Final new
Score (%)
λ_R
0.05
0.1
0.2


(a) OBJECT | S − C (matched)
(b) OBJECT | S − Q
(c) GOAL | S − C (matched)
(d) GOAL | S − Q
Training stage
Task
6
7
8
9
0.0
+18.3
-5.0
+25.0
-1.7
+3.3
+21.7
0.0
+3.3
+5.0
+5.0
0.0
-1.7
+6.7
+10.0
+15.0
+11.7
−20
−10
0
10
20
Success-rate difference (percentage points)


Seed 41
Seed 42
Seed 43
Mean
(a) OBJECT | FWT
(b) OBJECT | NBT
(c) GOAL | FWT
(d) GOAL | NBT
Score (%)
NBT (pp; lower is better)
100
90
80
70
60
20
10
0
−10
−20
0.05
0.1
0.2
λ_R